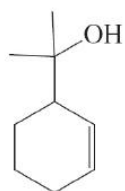
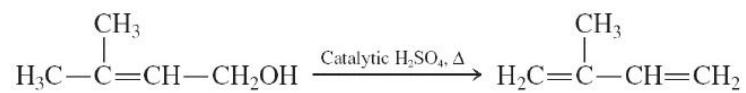


44. Starting with cyclohexene, propose a reasonable synthesis of the cyclohexene derivative shown below.



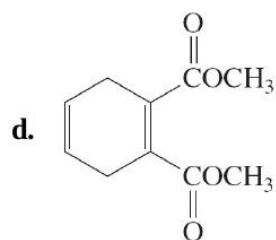
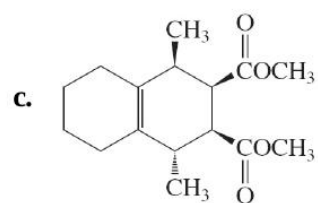
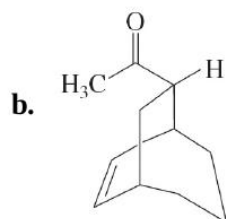
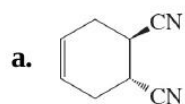
55. Sketch the molecular orbitals for the pentadienyl system in order of ascending energy (see [Figures 14-2](#) and [14-7](#)). Indicate how many electrons are present, and in which orbitals, for **(a)** the radical; **(b)** the cation; **(c)** the anion (see [Figures 14-3](#) and [14-7](#)). Draw all reasonable resonance forms for any one of these three species.

56. Dienes may be prepared by elimination reactions of substituted allylic compounds. For example,



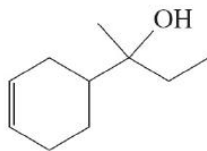
57. Give the structures of all possible products of the acid-catalyzed dehydration of vitamin A ([Section 14-7](#)).

58. Propose a synthesis of each of the following molecules by Diels-Alder reactions.

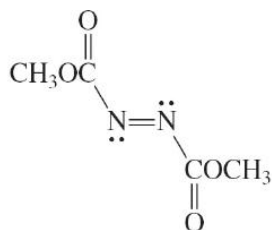


- 59.** The haloconduritols are members of a class of compounds called glycosidase inhibitors. These substances possess an array of intriguing biological functions from antidiabetic and antifungal to activity against HIV virus and cancer metastasis. Stereoisomeric mixtures of bromoconduritol (below) are commonly used in studies of these properties. One recent synthesis of these compounds proceeds via the

61. **CHALLENGE** Propose an efficient synthesis of the cyclohexenol below, beginning exclusively with acyclic starting materials and employing sound retrosynthetic analysis strategy. [**Hint:** A Diels-Alder reaction may be useful, but take note of structural features in dienes and dienophiles that permit Diels-Alder reactions to work well ([Section 14-8](#)).]

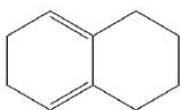


62. Dimethyl azodicarboxylate (see below) takes part in the Diels-Alder reaction as a dienophile. Write the structure of the product of cycloaddition of this molecule with each of the following dienes. **(a)** 1,3-Butadiene; **(b)** *trans, trans*-2,4-hexadiene; **(c)** 5,5-dimethoxycyclopentadiene; **(d)** 1,2-dimethylidenecyclohexane. Ignore the stereochemistry at nitrogen in the products (amines experience rapid inversion, as we shall see in [Section 21-2](#)).

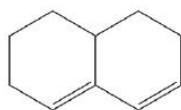


Dimethyl azodicarboxylate

- 63.** Bicyclic diene **A** reacts readily with appropriate alkenes by the Diels-Alder reaction, whereas diene **B** is totally unreactive. Explain.

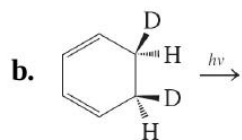
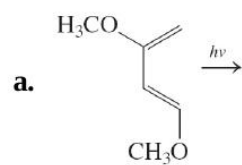


A



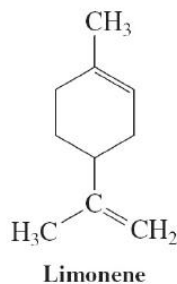
B

64. Formulate the expected product of each of the following conversions.



66. Explain the following reaction sequence. (**Hint:** Remember the Heck reaction, [Section 13-9](#).)

68. The structure of the terpene limonene is shown below (see also [Solved Exercise 5-29](#)). Identify the two 2-methyl-1,3-butadiene (isoprene) units in limonene. **(a)** Treatment of isoprene with catalytic amounts of acid leads to a variety of oligomeric products, one of which is limonene. Devise a detailed mechanism for the acid-catalyzed conversion of two molecules of isoprene into limonene. Take care to use sensible intermediates in each step. **(b)** Two molecules of isoprene may also be converted into limonene by a completely different mechanism, which takes place in the strict absence of catalysts of any kind. Describe this mechanism. What is the name of the reaction?

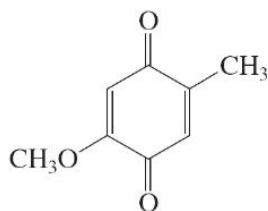


70. What is the longest-wavelength electronic transition in each of the following species? Use molecular-orbital designations, such as $n \rightarrow \pi^*$, $\pi \rightarrow \pi^*$, $\pi_1 \rightarrow \pi_2$, $\pi_1 \rightarrow \pi_2^*$, in your answer. (**Hint:** Construct a molecular-orbital energy diagram, like that in [Figure 14-16](#), for each.)
- (a) 2-Propenyl (allyl) cation; (b) 2-propenyl (allyl) radical; (c) formaldehyde, $\text{H}_2\text{C}=\text{O}$; (d) N_2 ; (e) pentadienyl anion ([Problem 55](#)); (f) 1,3,5-hexatriene.

71. Ethanol, methanol, and cyclohexane are commonly used solvents for UV spectroscopy because they do not absorb radiation of wavelength longer than 200 nm. Why not?

72. The ultraviolet spectrum of a $2 \times 10^{-4} \times 10^{-4}$ M solution of 3-penten-2-one exhibits a $\pi \rightarrow \pi^*$ absorption at 224 nm with $A = 1.95$ and an $n \rightarrow \pi^*$ band at 314 nm with $A = 0.008$. Calculate the molar absorptivities (extinction coefficients) for these bands.

76. Diels-Alder cycloaddition of 1,3-butadiene with the cyclic dienophile shown below takes place at only one of the two carbon-carbon double bonds in the latter to give a single product. Give its structure and explain your answer. Watch stereochemistry.



This transformation was the initial step in the total synthesis of cholesterol ([Section 4-7](#)), completed by R. B. Woodward (see [Section 14-9](#)) in 1951. This achievement, monumental for its time, revolutionized synthetic organic chemistry.

Team Problems

78. How many nodes are present in the LUMO (lowest unoccupied molecular orbital) of 1,3-butadiene?
- a. Zero;
 - b. one;
 - c. two;
 - d. three;
 - e. four

- 80.** When cyclopentadiene is treated with tetracyanoethene, a new product results. Its most likely structure is